

UNITED STATES MINERAL RESOURCES

COPPER

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ABSTRACT OF CONCLUSIONS

About 70 percent of the world's identified copper resources can be grouped in the following geologic-geographic categories, listed in order of decreasing importance: porphyry copper deposits of Chile and Peru, porphyry copper deposits of the Southwestern United States (19 percent of the world total), sedimentary copper deposits of Zaïre and Zambia, and porphyry copper and sedimentary copper deposits of the U.S.S.R. Other important groups of identified resources include porphyry copper deposits in Oceania, in Mexico, and in western Canada and porphyry copper and sedimentary copper deposits in southeastern and central Europe, respectively. These and the remaining identified resources amount to 344 million tons of copper metal, sufficient to last for 50 years at present rates of mine production.

Hypothetical resources, or those not now known but suspected to be present near known deposits, will probably provide another 400 million tons of copper. Important resources of this category are most likely to occur in the porphyry copper provinces of southeastern Arizona, western Canada, Chile, and Peru. Large sedimentary deposits will probably be found in Africa, the U.S.S.R., and the northern Rocky Mountains of the United States. Additional resources in sea-floor manganese nodules also will probably be discovered.

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It is possible to speculate that another 320 million tons of copper might be found in areas not yet prospected and in which important deposits are unknown. The most promising areas lie in the Basin and Range province of the United States, outside the known porphyry copper province; in the U.S.S.R. Far East where porphyry copper provinces may be found in Mesozoic fold belts and the younger island arc system; and in the Brazilian shield where major sedimentary copper deposits might be found.

The aforementioned identified, hypothetical, and speculative categories refer to known or postulated ore deposits, minable under present economic and technological conditions. In addition to these are large conditional resources in porphyry copper, sedimentary copper, and igneous copper-nickel deposits and in manganese nodules on the sea floor. At least 381 million tons of copper in such subeconomic deposits can be identified or hypothesized. This copper may become available at some future period of high metal prices and (or) improved technology.

A comparison of current U.S. and world production and the copper content of various copper deposits shows that the United States produces enough copper to mine out an average-sized porphyry copper deposit each year. Maintaining a balance between deposits discovered and deposits mined out means that a great investment in geological, geochemical, and geophysical studies must be made and maintained in spite of short-term fluctuations in the price of copper.

PROPERTIES, USES, AND RESOURCES

Copper, one of the most useful and versatile metals, has been used by man perhaps longer than any other metal except gold. Its unique physical and mechanical properties are utilized both in its metallic state and in alloy with many other metals. Copper, chemical symbol Cu, has a hardness (Mohs' scale) of 2.5-3, a melting point of 1,083°C, a boiling point of 2,595°C, a density (annealed at 20°C) of 8.89, and an electrical resistivity at 20°C of 1.71. Its electrical conductivity in cross section and by weight is surpassed only by silver and by aluminum, respectively.

High electrical and thermal conductivities, good resistance to corrosion, good ductility and malleability, high strength, lack of magnetism, and a pleasing red color are properties and characteristics of copper that find innumerable industrial applications. Copper and many of its alloys can be joined by welding, brazing, and soldering. It can be finished by plating and lacquering.

The four traditional copper-base alloys, brass, bronze, nickel-silver, and cupronickel, contain no less than 40 percent copper, with the amount of copper not less than that of any other constituent. Copper also is used in many other alloys where it is not the major component.

About half (53 percent) of the 1.5 million tons of copper used domestically in 1970 was for elec-

trical applications, followed by 16 percent in construction, 12 percent in industrial machinery, 8 percent in transportation, 6 percent in ordnance, and 5 percent in other uses.

Use of copper in all applications for which it is suited would exceed its available supply, but aluminum has substituted for copper in many applications. In the last two decades U.S. consumption of aluminum has more than tripled, while the use of copper increased only 15 percent. Although copper is virtually indispensable for some applications, aluminum serves satisfactorily as an alternate metal in many others, and often at lower cost. The relative capabilities of the two metals in their electrical conductor applications depend on their physical and mechanical properties. For example, an aluminum conductor weighs only 30 percent as much as a copper conductor of identical size, but an aluminum conductor has 1.64 times the electrical resistance of a copper conductor of equal size. Thus, for equal current-carrying capacity, an aluminum conductor weighs half as much as a copper conductor, but has a cross-sectional area 26 percent greater.

Aluminum and stainless steel are used in place of copper for some applications in the building industry. Steel increasingly replaces brass in shell cases. The use of copper cladding proportionately requires less copper than would solid copper; copper now substitutes for silver in coins which have a copper core and a cupronickel cladding material. Printed electrical circuits and plastics have replaced copper in many applications. These shifts in the overall use pattern make more copper available for preferred uses. In spite of substitution, the continued use of a substantial part of our present copper requirement is needed to maintain or present living standard.

The supply to meet domestic demand is derived from U.S. mine production, secondary production (recovery of old scrap), imports, and withdrawals from industrial and Government stockpiles. The trends in supply and demand for copper during the period 1950-70 are illustrated in figure 20 and table 36. Other demand forecasts to the year 2000 are discussed by Lowell (1970) and by Ageton and Greenspoon (1970).

Copper is traded in a number of commercial classes—ore, concentrate, matte, blister, refined, and secondary—each with varying percentages of copper content. Because of the complexity in determining the total amount of copper used from all classes and the many demands of trade such as imports, exports, and stockpiles, the total demand

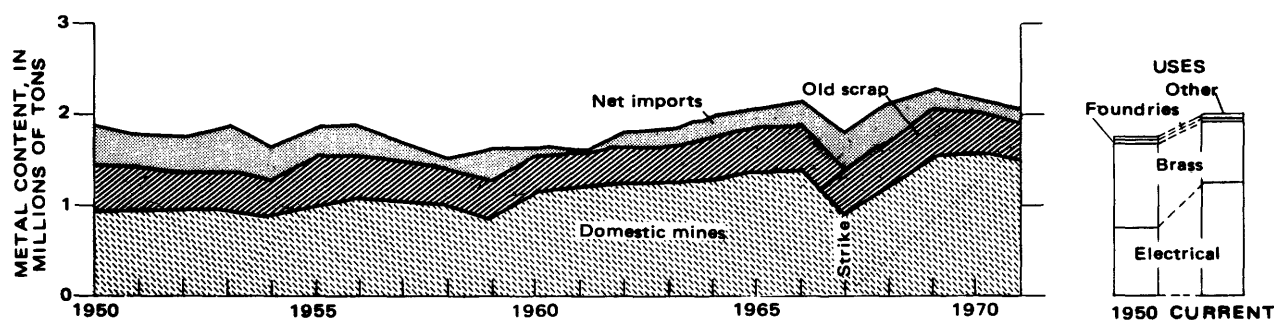


FIGURE 20.—U.S. supplies and uses of copper, 1950-71. (From U.S. Department of the Interior (1972).)

TABLE 36.—Copper supply and demand data, 1950-71, and projections for the year 2000, in thousands of short tons

[Data from National Commission on Materials Policy (1972), U.S. Department of the Interior (1972), and U.S. Bureau of Mines (1970)]

Year	Refined production U.S. mines	Old scrap	U S net imports	Total U.S. demand, exclusive of new scrap	Rest of world mine production
1950	921	485	491	1,897	1,851
1960	1,121	429	107	1,657	3,570
1970	1,521	504	155	2,180	4,847
1971 ¹	1,400	450	190	2,040	(²)
2000 ³	2,380	1,700	(²)	9,700	(²)

¹ Preliminary.

² Not available or estimated.

³ Unrefined.

often is expressed as withdrawals or apparent consumption.

With few exceptions, the United States has been a net importer of copper each year since 1939. Dependency on foreign sources of supply has ranged from a high of about 60 percent in 1945 and 1946 to 6 percent in 1971.

The major sources of copper imports in 1971 were Canada, 37 percent; Peru, 29 percent; Chile, 15 percent; and all other countries, 19 percent. Major destinations of copper exports in 1971 were West Germany, 26 percent; Italy, 11 percent; France, 10 percent; Canada, 9 percent; and all other countries, 44 percent.

The sources of 1971 imports differ markedly from the sources for the 1967-70 period, when 27 percent originated in Chile, 23 percent in Canada, 21 percent in Peru, 7 percent in the Republic of South Africa, and 22 per from all other countries. The nationalization in 1971 of mines owned by U.S. firms in Chile disrupted world trade patterns in copper, as is evident from the preceding data.

The possible effect of industrialization of developing countries on the world copper market is best illustrated by the Peoples Republic of China. Present per-capita copper consumption in China is about one-half pound per year, and many recent studies of the world metal situation have pointed to the

impact of the increase that is sure to come with advancing industrialization in that and other populous, mostly agricultural, countries. Even though Chinese copper production may increase rapidly, consumption may well outrun it for many years, and China probably will become a major importer in the short term, though self-sufficiency may be possible at a future time. The 4-year agreement for Chinese purchase of copper from Chile is an example of this kind of change that is sure to have widespread effect on the world copper market.

EXPLOITATION

The introduction of new domestic sources of copper, important new developments, and other events having significant impact on the supply of copper are shown in relation to primary copper production in figure 21.

Development of foreign deposits also has had its impact on the copper industry. The more economically important developments were the following: 1912, Braden Copper Co. began operations in Chile. 1916, Chile Copper Co. started production at Chuquibambilla.

1924, Katanga in Zaïre (Congo) started production. 1950, 1960's, Large deposits developed in Chile and Peru. Substantial production started and reserves added in Canada, Zambia, Zaïre, Yugoslavia, Poland, Ireland, Finland, Bulgaria, U.S.S.R., Australia, and the Philippines.

Among the foreign developments begun in the 1960's and expected to be in production in the 1970's are those in Indonesia (West Irian), New Guinea (Manus Island, Bougainville), British Columbia, and Iran.

Two factors of paramount importance to the copper industry are the grade of ore mined and the price per pound. The steady decline in average grade of domestic ore mined and the wide range in prices are evident in the following summary:

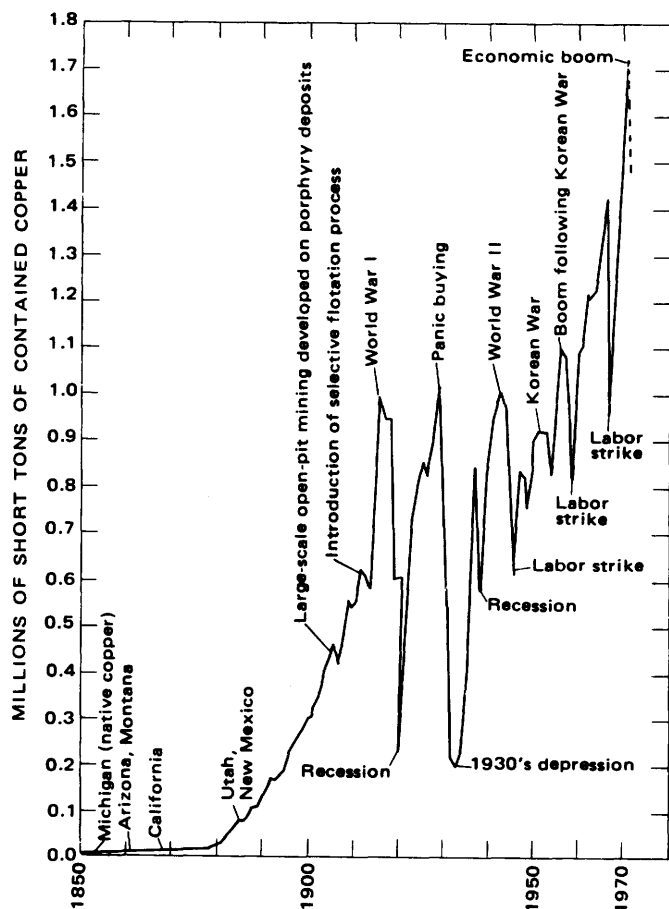


FIGURE 21.—U.S. primary copper production, 1850–1970.
Notes indicate major new sources, developments, or events.

Year	Average annual price in cents per pound		Average grade of ore mined (percent copper)
	Quoted refinery	Equivalent 1970 dollars	
1900	16.19	63.5	4.0
1910	12.73	43.5	1.88
1920	17.45	30.0	1.63
1930	12.98	32.5	1.43
1940	11.29	29.5	1.20
1950	21.32	30.0	.89
1955	37.49	41.0	.83
1960	32.05	36.5	.73
1965	35.02	39.9	.70
1970	58.20	58.2	.60

Copper ores also contain other valuable recoverable metals, and many ores mined chiefly for other metals also contain recoverable copper. The coproduct-byproduct recovery of copper and associated metals from ores depends upon the quantity of metals present (grade) and the processes used in recovering the principal values. Copper production yields significant quantities of gold, silver, molybdenum, cobalt, zinc, lead, nickel, sulfur, tellurium, selenium, rhenium, palladium, platinum, arsenic, and iron. Selenium is an essential byproduct of copper production, and rhenium is obtained from the

molybdenum in porphyry deposits. Cobalt is derived mainly from copper ores mined in Zaïre.

Several problems of major concern to copper producers relate to protecting environmental qualities while maintaining competitive costs. Copper smelters are under severe new or proposed restrictions on sulfur emission, requiring major capital investment for control equipment. This stress may result in the closing of some smelters and a shift to new methods of hydrometallurgy. Surface restoration is particularly difficult in copper mining because of the finely powdered state of mill tailings and the very high ratio of waste to produced metal. Water pollution by mine and mill waste is a problem, and particularly as many of the big copper deposits are in arid regions, obtaining adequate water supplies for mining and milling becomes more difficult.

Several porphyry copper deposits in Arizona are now being worked by in-place leaching methods which greatly decrease the effect of copper extraction on the environment. The deposits are first cut by a system of tunnels, then fractured by explosive or hydraulic methods, and then leached by sulfuric acid introduced through pipes to the upper part of the deposit. The copper-rich fluids are withdrawn from the lowest levels of mine workings, pumped to the surface and the copper removed electrolytically. The main disadvantages of this method are that only 50–60 percent of the copper in the ore deposit can be recovered under present techniques, and that all of the byproduct metals, such as molybdenum, gold, and selenium, are left in the rock during the leaching process.

GEOLOGIC ENVIRONMENT

GEOCHEMISTRY AND ABUNDANCE

Copper is a chemical element of group II-B of the periodic table; it has atomic number 29 and atomic weight 63.546, consisting of the isotopes ^{63}Cu (69.09 percent) and ^{65}Cu (30.91 percent). It occurs in nature as the native element, as compounds of Cu^{+1} (ionic radius 0.54 Å) and Cu^{+2} (ionic radius 0.81 Å for sixfold coordination). In the Cu^{+2} state, as shown particularly by naturally occurring sulfates and phosphates, it is isomorphous with Zn (0.83 Å), Mg (0.80 Å), and Fe^{+2} (0.69 Å). In sulfides, Cu^{+2} is isomorphous with zinc (as in tetrahedrite), and Cu^{+1} with silver. Copper is a strongly sulfur-seeking (chalcophile) element, and it tends to be concentrated in sulfide deposits; in igneous rocks it occurs mainly as finely divided sulfides, principally chalcopyrite and bornite.

Estimates of the abundance of copper in the con-

tinental crust have been reviewed by Parker (1967); most recent estimates are about 50 ppm (parts per million). As indicated in table 37, copper is con-

TABLE 37.—*Copper content of various materials*

Type of material	Copper content generally reported	
	Range	Average
Igneous rocks (parts per million)		
Ultramafic rocks -----	-----	15
Basaltic and gabbroic rocks -----	30-160	90
Andesites -----	-----	35
Granitic rocks -----	5-30	15
Sedimentary rocks (parts per million)		
Limestones -----	-----	4
Sandstones -----	¹ 10-20	10 (?)
Shales and clays -----	18-120	² 45
Soils -----	2-100	20
Phosphorites -----	10-100	30
Coals -----	2-40	15
Waters (parts per billion)		
Sea Water -----	1-15	3
Surface waters -----	.4-150	5 (?)

¹ Pettijohn (1963).

² 250 for deep-sea clays (Turekian and Wedepohl, 1961); 560 average for 450 samples of deep-sea clay (Landergrén and Manheim, 1963).

centrated in the igneous cycle in basaltic and gabbroic rocks. In these it is highest in the ferromagnesian minerals such as pyroxenes and biotite where it probably occurs mainly as minute grains of chalcopyrite (Goldschmidt, 1954, p. 182).

During the crystallization of igneous rocks, copper and some other elements that do not fit readily into the structures of silicate minerals may be concentrated to form ore deposits. If the magma contains appreciable volatiles, copper may be dissolved and transported with these into veinlets or extensive vein systems. If, as in the case of some gabbroic rocks, the volatile content is low, copper may remain in the intrusion, concentrating as an immiscible copper sulfide fluid.

The data on sea water indicate that less than 0.1 percent of the copper carried to the ocean remains in solution, the remainder being precipitated—mainly with the clays, in part with manganese oxides, and in part removed by biota. Probably much, perhaps most, of the copper reported in surface waters is due to industrial contamination from metallurgical wastes, plating work, and so forth.

Copper is found in at least trace amounts in nearly all sedimentary rocks. The averages for some common rocks are shown in table 37. The copper content of Lower Eocene nonmarine sandstones in the Rocky Mountain region was found to average 11 ppm and, for individual basins, to range from 5.2 to 29 ppm (Vine and Tourtelot, 1970a). Pelitic Belt rocks from the Mission Mountains Primitive Area, Mont., contain an average of 14 ppm copper,

whereas similar rocks in the Pend Oreille, Idaho, area contain 6.8 ppm copper (Harrison and Grimes, 1970, table 11).

Copper is readily soluble and mobilized by oxidizing solutions; therefore, it concentrates in certain environments of deposition. Thus, marine clays and other fine-grained rocks tend to have more copper than coarse-grained rocks and limestones. Moreover, the average black shale contains about twice as much copper as the average shale, and some black shales rich in organic matter contain several hundred parts per million copper (Vine and Tourtelot, 1970b). Pelagic sediments contain significantly greater concentrations of copper than continental rocks. Landergrén and Manheim (1963) reported 560 ppm copper as the average for 450 samples of deep-sea clay. Bostrom and Peterson (1966) found more than 1,000 ppm copper and an enrichment of several heavy metals along the crest of the East Pacific rise. A metal-rich sediment associated with hot brine from the Atlantis II deep in the Red Sea is reported to contain 0.5 percent copper (Miller and others, 1966).

GEOCHEMICAL CYCLE

Copper is introduced into the accessible part of the earth's crust from unknown deeper sources by igneous intrusions and by upward migrating fluids. Economic concentrations of the metal may result directly from these processes and from secondary effects of weathering, erosion, and sedimentation. Remobilization and reconcentration by numerous later stages of intrusion and by metamorphism may occur, and in addition ground water and deep connate waters may dissolve, transport, and redeposit copper to form younger ore deposits. Copper in oceanic basalt and copper carried to the sea in solution and deposited with oceanic sediments may be remobilized during subduction of oceanic crustal plates, and a part may ultimately be redeposited in continental crustal rocks above the subduction zone.

The weathering of chalcopyrite readily oxidizes copper and releases it to ionic solution. A portion of this ionic copper may be adsorbed on organic matter, clay minerals, and iron oxide particles, and some of it probably reaches the sea in ionic form. According to Robert Ehrlich and T. A. Vogel (oral commun., 1972), the absence of much potassium is an important factor in the transport of copper as a complex with montmorillonite. The copper in solution in river waters ranges widely, but 0.010–0.015 ppm is probably the general range for large rivers. Some copper is believed to have been transported to the Siberian Udokan deposits in detrital form

(Bogdanov, 1968), and Ehrlich and Vogel (oral commun., 1972) suggested that some of the copper in the Nonesuch Shale was carried in this way.

Because sea water contains only 0.003 ppm copper (Green, 1969), the discharge of an estimated 1×10^6 tons new copper by streams each year must be balanced against various mechanisms that remove it. In reducing environments, copper is probably precipitated as the sulfide by sulfur-reducing bacteria or adsorbed on decaying humic organic matter. Copper also accumulates in some oxidizing environments, such as in manganese nodules and in pelagic muds where precipitation may occur as the oxide. The fate of copper in an evaporite environment is not understood. Some copper may be concentrated in residual brines and eventually squeezed into adjacent porous sediments and precipitated by reaction with hydrogen sulfide. Copper may also be trapped in connate pore waters in basins of nonmarine deposition. Precipitation would eventually occur by reaction with hydrogen sulfide or organic matter, or as a carbonate, sulfate, silicate, or phosphate. The regional and stratigraphic association of stratabound copper ores with evaporites was described by Davidson (1965).

Prior to the evolution of a modern atmosphere containing about 20 percent oxygen some 1.2-1.4

billion years ago (Cloud, 1971), the geochemical cycle for copper may have been significantly different. Primary copper sulfide minerals were probably sufficiently stable to survive weathering and redeposition as heavy minerals in detrital rocks. Subsequent exposure of buried sediment to oxygenated waters could have resulted in redistribution of the copper in chemically reactive environments within the sedimentary sequence, for example in beds rich in organic matter.

MINERALOGY

Copper occurs in at least 160 minerals. Table 38 gives the name, composition, and general mode of occurrence of 20 of the more common ones.

Chalcopyrite is by far the most abundant copper sulfide; bornite and chalcocite are next in abundance. The sulfarsenides enargite and tennantite and the sulfantimonides tetrahedrite and farnatinitite are generally rare, but each makes up major parts of at least one large ore body. The mineral digenite ($\text{Cu}_2\text{-S}$) is not shown in table 38, but it may be a major ore mineral. It is detected only by careful petrographic work or X-ray analysis and is generally reported as chalcocite. Native copper is abundant in certain types of deposits.

Malachite, azurite, and chrysocolla are the most

TABLE 38.—*Minerals of copper; composition and environment of origin*

[o, major ore mineral; x, minor ore mineral]

Name	Composition	Porphyry	Vein and replacement	Sedimentary strata-bound	Volcanogenic strata-bound	Magmatic segregation	Keweenaw type	Secondary enrichments and secondary deposits	
								Oxide zone	Sulfide zone
Major minerals									
Native copper -----	Cu -----	--	--	o	--	--	o	x	x
Chalcocite -----	Cu ₂ S -----	--	o	o	o	--	x	--	o
Covellite -----	CuS -----	--	--	--	--	--	--	--	o
Bornite -----	Cu ₅ FeS ₄ -----	x	o	x	o	o	--	--	--
Chalcopyrite -----	CuFeS ₂ -----	o	o	o	o	o	--	--	--
Enargite -----	Cu ₃ AsS ₄ -----	o	o	--	x	--	--	--	--
Cuprite -----	Cu ₂ O -----	--	--	--	--	--	--	o	x
Malachite -----	Cu ₂ (OH) ₂ (CO ₃) -----	--	--	--	--	--	--	o	--
Significant supplementary minerals									
Tetrahedrite -----	(Cu,Fe) ₁₂ Sb ₄ S ₁₃ -----	--	x	--	x	--	--	--	--
Tennantite -----	(Cu,Fe) ₁₂ As ₄ S ₁₃ -----	--	x	--	x	--	--	--	--
Famatinite -----	Cu ₃ SbS ₄ -----	--	x	--	--	--	--	--	--
Stannite -----	Cu ₂ FeSnS ₄ -----	--	x	--	(?)	--	--	--	--
Atacamite -----	Cu ₂ (OH) ₂ Cl -----	--	--	--	--	--	--	x	--
Tenorite (melaconite) -----	CuO -----	--	--	--	--	--	--	o	--
Azurite -----	Cu ₂ (OH) ₂ (CO ₃) ₂ -----	--	--	--	--	--	--	x	--
Chrysocolla -----	CuSiO ₃ •2H ₂ O -----	--	--	--	--	--	--	o	--
Brochantite -----	Cu ₄ (SO ₄)(OH) ₆ -----	--	--	--	--	--	--	x	--
Antlerite -----	Cu ₃ (SO ₄)(OH) ₄ -----	--	--	--	--	--	--	o	--
Chalcantithite -----	CuSO ₄ •5H ₂ O -----	--	--	--	--	--	--	o	--
Kroehnkite -----	Na ₂ Cu(SO ₄) ₂ •2H ₂ O -----	--	--	--	--	--	--	x	--

common minerals of the oxidized zone, and the bright colors of these minerals have been a guide to prospectors for thousands of years.

TYPES OF DEPOSITS

Most copper deposits fall into one of five major types. (1) The most important of these are the porphyry copper deposits and vein and replacement deposits; these account for almost two-thirds of the world's copper resources and are discussed together because of their common genetic association with felsic intrusive rocks. (2) A fourth of the world's identifiable copper resources is in sedimentary rocks, and (3) about 5 percent is in volcanic rocks as massive sulfide deposits. (4) Smaller portions of world resources are in nickel-copper ores formed by magmatic processes in mafic intrusives and (5) in native copper ores of the Keweenaw type. These types, as well as certain uncommon or unique types of deposits, are discussed in order of their relative importance as copper resources.

PORPHYRY COPPER DEPOSITS AND RELATED TYPES

Porphyry copper deposits and most of the important vein and replacement copper deposits are genetically associated with intrusions of felsic igneous rocks. The copper in these deposits is believed by some geologists (for example, Fournier, 1967) to be a primary constituent of the associated intrusion. A part of the copper is trapped in disseminated grains by the rapid crystallization of the magma, which in turn gives rise to the characteristic porphyritic texture of the intrusion. Another part of the copper is mobilized by water and other volatiles escaping from the hot congealed rock mass and is deposited in fractures in the intrusion and its wall rocks. A third part may escape completely from the intrusion and form vein and replacement deposits in nearby reactive host rocks. Thus porphyry copper deposits commonly occupy the central part of large or small mining districts containing vein and replacement deposits of copper, lead, zinc, silver, gold, iron, and manganese.

PORPHYRY COPPER DEPOSITS

An excellent review of the geology of porphyry copper deposits of the North American Southwest with detailed descriptions of 15 individual deposits was assembled by Titley and Hicks (1966). Sutherland-Brown (1969) and Sutherland-Brown, Cathro, Panteleyev, and Ney (1971) wrote excellent descriptions of deposits in British Columbia.

Porphyry copper deposits are defined herein as deposits of disseminated copper sulfide that are in or near a felsic intrusive body; in most deposits, at

least one facies of the intrusive is a porphyry. This definition does not include size, grade, or amenability to mass-mining methods because these factors restrict consideration of the similar but subeconomic occurrences that are so important to regional metallogenic studies. Lowell and Guilbert (1970) tabulated the geologic features of 27 major porphyry deposits, 18 of them in the North American Southwest, and arrived at a typical or average deposit. This hypothetical ore body is oval and pipe-like, it measures 3,500 by 6,000 feet in plan, and it contains 150 million tons of ore averaging 0.8 percent copper and 0.15 percent molybdenum. Ore below the zone of secondary enrichment contains on the average only 0.45 percent copper. On the average 70 percent of the ore body is in the intrusion, and 30 percent is in the country rock. Sulfide minerals in descending order of abundance are pyrite, chalcopyrite, molybdenite, and bornite. Supergene sulfides may form a higher grade zone of secondary enrichment.

Porphyry copper deposits have petrologic associations that are dependent on their tectonic environment. Deposits formed on thick continental crust are usually associated with quartz monzonite, as in Arizona (Lowell and Guilbert, 1970). Deposits in island arcs are associated with quartz diorite, diorite, or granodiorite, as in Bougainville (Macnamara, 1968), the Philippines (Madamba, 1972; Loudon, 1972), and Puerto Rico (Cox, 1971). Similarly, British Columbia porphyry deposits of two distinct ages and of distinct petrologic affinities seem to be related to two types of crustal plate motion (Hodder and Hollister, 1972). Thus, deposits of Triassic and Jurassic age were formed during plate convergence as the North American plate moved westward over the Pacific plate; as the East Pacific rise drifted under the American plate, development of tensional structures in the continental crust and a new regimen of magmatic activity gave rise to the Cretaceous and Tertiary deposits. The deposits formed in the earlier cycle in a thin or poorly developed continental crust are associated either with syenite, monzonite, or fennite as at Stikkeen and Copper Mountain or with quartz diorite and granodiorite as at Highland Valley. The later deposits that formed in a thicker continental crust are associated with quartz monzonite.

Breccia pipes are present in 20 of the 27 deposits considered by Lowell and Guilbert (1970), and wide zones of closely and irregularly fractured rock (crackle zones) surround almost all the deposits. This last feature is a very useful one in the recognition of porphyry deposits in the field.

Lowell and Guilbert (1970, p. 403–404) described the hydrothermal alteration of a typical deposit as forming in response to a “physical-geochemical continuum from low-temperature magmatic to ‘conventional’ hydrothermal conditions” in which hydrogen ion concentrations increase outward or with time. Where the country rock is homogeneous, a series of symmetrical, concentric shells of distinct hydrothermal mineral assemblages may be produced around the igneous intrusive. From the interior of the deposit outward, these are the potassic (quartz-K-feldspar-biotite), the phyllic (quartz-sericite-pyrite), the argillic (quartz-kaolin-chloride), and the propylitic (chlorite-epidote-carbonate-adularia and albite).

Many factors limit or disrupt the symmetry of the model porphyry copper deposit. Lowell and Guilbert listed (1) a strong regional structural fabric, (2) heterogeneity of preore host rocks, (3) postore faulting and intrusion, and (4) different levels of erosion exposing the deposits.

Titley (1972) drew attention to the profound effect of the country rock on the alteration mineralogy and on the form of the ore bodies. He distinguished “intrusion porphyry copper deposits” in which the ore body is mostly within the intrusive porphyry, from “wall rock porphyry copper deposits” in which the wallrock near a porphyry intrusion is altered and mineralized. Composites of these types are common. Titley emphasized that ore bodies in wallrocks may be just as valuable as those in intrusions and should not be overlooked because their alteration pattern does not follow the ideal model.

Porphyry copper deposits of the Pacific and Caribbean islands exhibit alteration patterns that diverge sharply from the Lowell and Guilbert model. Biotite is the most common mineral in the potassic alteration facies, and K-feldspar is commonly lacking, possibly because of the low availability of potassium in the intrusive quartz diorite and diorite porphyries and surrounding andesitic volcanic rocks. Propylitic alteration is widespread in these deposits, and in Puerto Rico it appears to follow an early period of biotitic alteration (Cox, 1971).

Figure 22 shows the size and grade characteristics of various porphyry copper deposits. The Bingham deposit in Utah is the largest in the United States, with a production since 1904 of over 14 million tons of copper (Anaconda Copper Corp., annual reports). The field of 22 U.S. deposits in production in 1960 has its center point at about 100 million tons at 0.8 percent copper or about 800,000 tons of copper metal per deposit.

A second field of eight U.S. and Canadian deposits brought into production in the last 10 years shows a downward trend in grade. At Sierrita, Ariz., the mining of ore averaging 0.34 percent copper is made possible by lower cost mining and processing methods. These new deposits contain on the average about 2 million tons of copper each. Likewise, porphyry deposits with byproduct molybdenum discovered in other parts of the world are being brought into production at average copper grades as low as 0.18 percent (Brenda, British Columbia) and 0.3 percent (Casino, British Columbia) (Peters, 1970). La Caridad, Mexico (Peters, 1970) and the combined San Manuel-Kalamazoo deposits, Arizona (Peters, 1970; Thomas, 1966, p. 139) are the largest of the new deposits, with about 7 million tons of copper each.

Overshadowing these are the great Andean porphyry deposits of Chile and Peru. These deposits were brought into production with average grades of 1–2 percent, and the average of 11 deposits plotted is about 4 million tons of copper each. The El Teniente and Chuquicamata deposits in Chile, which would lie on the right side of this field in figure 22, are larger by at least an order of magnitude than the majority of North American porphyry deposits. It is obvious that the Andes represent a copper province of great wealth and that numerous other lower grade deposits await development.

The principal byproduct metals of porphyry deposits are molybdenum, gold, and silver. King (see “Molybdenum” chapter) gives a range of MoS_2 values of 0.015–0.1 percent for porphyry copper deposits; Simons and Prinz (see “Gold” chapter) give a range of 0.001–0.025 ounce gold per ton in copper deposits in general. An average silver value of 0.06 oz per ton is given for 14 Arizona porphyry deposits by Heyl and others (see “Silver” chapter). Kesler (1971) pointed out that continental porphyry deposits have higher molybdenum values and that island arc deposits have higher gold values and lower molybdenum values. Gold content of the Panguna deposit in Bougainville is 0.018 oz per ton; the Marcopper deposit in the Philippines contains 0.001 oz per ton (Espie, 1971; Loudon, 1972). Molybdenum values in these deposits are negligible.

Economically important trace amounts of selenium, tellurium, and rhenium are derived from copper and molybdenum concentrates from porphyry deposits. Pyrite concentrates are recovered for their use in sulfuric acid manufacture, and magnetite concentrates are recovered from some porphyry deposits such as Atlas in the Philippines (Madamba, 1972).

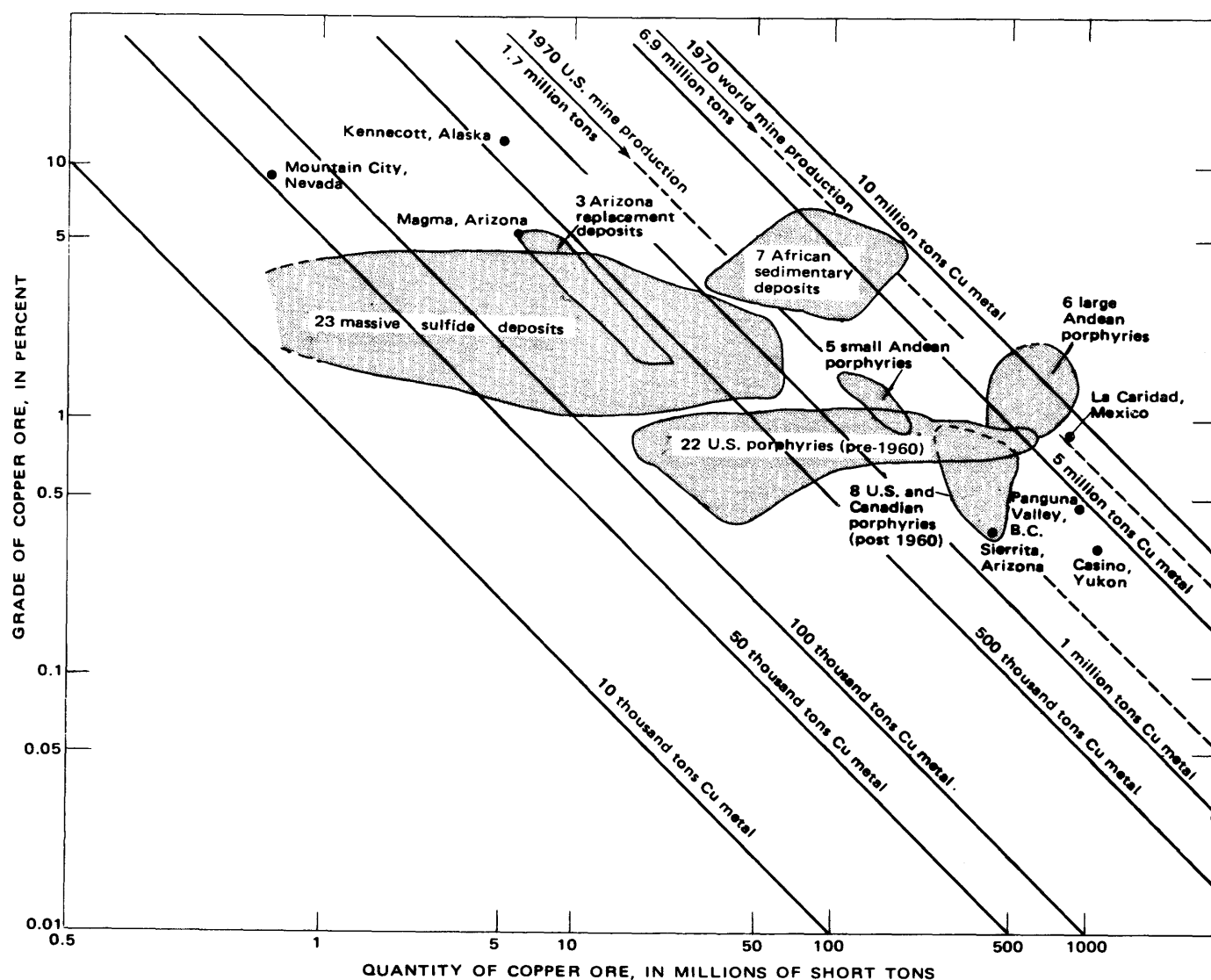


FIGURE 22.—Size and grade characteristics of selected copper ore bodies. Diagonal lines show tons of copper metal per deposit; world and U.S. mine production of copper metal are shown for comparison by diagonal dashed lines. Data from unpublished compilation by A. R. Kinkel, Jr., and others and from Peters (1970).

VEINS, PIPES, AND REPLACEMENT DEPOSITS

Copper-rich veins, pipes, and replacement deposits may be localized in felsic plutonic rocks with local porphyry intrusions as at Butte, Mont. (Meyer and others, 1968); in favorable host rocks near a porphyry copper deposit as at Bingham, Utah, and Bisbee, Ariz.; or near barren felsic intrusive rocks as at Magma, Ariz. (Hammer and Peterson, 1968) or Mission, Ariz. (Kinnison, 1966). Veins are formed when metal-rich solutions, emanating from a crystallizing intrusion, deposit minerals in faults or fractures. They are tabular in form and may show mineral textures indicating growth of crystals in open cavities as well as replacement of wallrocks

by copper minerals. Many veins show several generations of mineral deposition related to opening and reopening of fault zones by tectonic movements.

Hydrothermal alteration envelopes may accompany the veins, indicating chemical reaction between wallrocks and mineral-rich solutions. Sales and Meyer (1949), in their monumental study of quartz-sulfide veins at Butte, Mont., described a sequence of alteration zones outward from the veins into the quartz monzonite wallrocks. The sequence is consistently sericite to kaolin to montmorillonite to fresh wallrock. Comparison of the chemical composition of these zones suggests diffusion of some ions from the rock into the vein and simultaneous migration of hydrogen ions into the wallrocks. There

is a close similarity in mineralogy between large-scale alteration zones extending outward from porphyry deposits and small-scale alteration zones successively enveloping sulfide veins.

Ore pipes are rodlike in form as contrasted to tabular veins. They may form in fractured or brecciated rock at the intersections of faults or by explosive release of gases derived from intrusive magmas. These latter are called breccia pipes because of the broken rock associated with them. At many places copper deposits in breccia pipes show a close spatial and genetic relation to mineralized porphyry intrusions as at Cananea, Mexico (Velasco, 1966) or a close geographic relation to porphyry copper deposits as in the porphyry copper belt in Chile (Sillitoe and Sawkins, 1971).

Replacement deposits form in reactive host rocks near the intrusive contacts or along mineral veins. These hosts may be sedimentary rocks such as limestone, dolomite, or calcareous sandstone or even diabase sills as at Ray, Ariz. (Metz and Rose, 1966). Replacement deposits tend to be tabular in form and may be roughly concordant with bedding. Though high in grade at some mines, when these deposits are removed as part of a large low-grade ore body they are generally regarded as part of the porphyry copper ore body, as at the Mission and Pima mines in Arizona. Mining of porphyry ore at Bingham, Utah, encroaches steadily on chalcoppyrite replacement bodies in limestone beds to the west, and the distinction between ore in porphyry and ore in wallrocks becomes academic.

Favorable host rocks for replacement deposits may have continuity over large areas, and the recognition of their stratigraphic position is of great importance in the search for new deposits. In Arizona, increasing attention is being paid to details of local stratigraphy in attempts to locate hidden replacement deposits (Eyde, 1972).

Vein, pipe, and replacement ores have a more varied mineralogy than the porphyry deposits. Chalcoppyrite and bornite are the most common hypogene minerals. Chalcocite and covellite may be present as hypogene minerals in addition to forming secondary enrichment blankets. At Butte, Mont. (Meyer and others, 1968), main-stage mineralization gave rise to veins of four distinct types: veins of the deep-level chalcoppyrite zone containing chalcoppyrite, bornite, and tennantite; veins containing chalcocite-enargite in the central zone; bornite-chalcocite-chalcoppyrite-enargite veins in the intermediate zone; and carbonate veins containing rhodochrosite, sphalerite, galena, and rhodonite in the peripheral zone.

At the Magma mine in Arizona (Hammer and

Peterson, 1968) vein ores contain chalcoppyrite, bornite, enargite, tennantite, chalcocite, digenite, and sphalerite. Enargite predominates in the lower levels and gives way upward to tennantite, but sphalerite predominates in the upper levels. Digenite is present only in the lower levels. In the limestone replacement ores, chalcoppyrite is predominant.

Mineral zoning such as is exhibited at Butte and Magma is present to some extent in most vein and replacement deposits. In addition to this a larger scale mineral zoning has been noted around most mineral deposits related to felsic intrusive rocks. Lowell and Guilbert (1970) in their tabulation of 27 porphyry copper deposits described some peripheral mineralization in all the deposits and arcuate clusters of small mines and prospects in 23 of the deposits. Most of these peripheral veins contain gold and silver values along with pyrite, chalcoppyrite, sphalerite, specularite, enargite, famatinite, tetrahedrite, barite, and manganese and vanadium minerals. Recognition of a cluster of small veins as part of an outer zone surrounding a buried ore deposit may be an important step in the discovery of new copper resources.

Vein and replacement deposits are difficult to characterize in terms of tonnage and grade. If the Butte district is considered as one deposit—and no criteria seem to exist as a basis on which to subdivide it into smaller deposits—the amount of contained copper metal is enormous, rivaling that of the largest Andean porphyry deposits. In the period 1880 to 1964 the Butte mines produced more than 8 million tons of copper (Meyer and others, 1968). In Arizona, major vein and replacement deposits fall into a field (in fig. 22) with an average copper content of 350,000 tons of copper each.

In general, copper resources in vein, pipe, and replacement deposits are small in comparison to those of the closely related porphyry copper deposits, and discoveries of new deposits similar to Butte or Magma have declined sharply in the last few decades. The relatively small size of the ore bodies and alteration envelopes makes these deposits difficult targets in areas of deep overburden. It is likely, however, that vein, pipe, and replacement ore bodies will be discovered incidentally in the search for porphyry deposits.

STRATA-BOUND DEPOSITS IN SEDIMENTARY ROCKS

Strata-bound deposits in sedimentary rocks include some of the world's largest sources of copper. The sedimentary deposits of Zaïre and Zambia alone make up 15 percent (table 39) of the world's identified resources, and enormous deposits are known in

the U.S.S.R. Deposits of this type represent the largest potential for discovery of new deposits. Not only are many of them difficult to recognize in the field, but the geologic controls of their origin are only beginning to be understood.

A genetic classification of sedimentary copper deposits should distinguish ores formed by chemical or biological processes from those formed by mechanical processes; those deposited simultaneously with the enclosing sediment from those formed by later alteration; and those formed in a reducing environment from those formed in an oxidizing environment. However, these processes are not mutually exclusive, and the criteria for distinguishing them are frequently equivocal. Idealized descriptions of different types of sedimentary copper deposits presented below, followed by descriptions of several important deposits that may be combinations of types.

Type 1.—Rocks of intermediate Precambrian age formed when atmospheric oxygen began to become abundant and to mobilize copper that was previously stable. These deposits may include copper reworked or leached out of earlier disseminated deposits. The deposits of the African copper belt and those in the Belt Supergroup of western Montana may be of this type.

Type 2.—Syngenetic marine deposits formed on the sea floor by precipitation or adsorption on organic matter. These deposits should show a systematic relation to lithologic facies and may be characterized by lateral persistence within a bed or facies. Mineral textures are typical of sedimentary rocks and the deposits may contain other metals, such as uranium, vanadium, and silver, that could precipitate under similar conditions. The Kupferschiefer and equivalent formations in Europe may be of this type.

Type 3.—Deposits formed by reaction of connate brines or meteoric ground water with chemically reactive strata such as black shale, sandstone containing carbonaceous matter, or carbonate rock. As discussed by Lindgren (1911), such deposits probably form during compaction and lithification and are localized by variations in permeability, structure, and chemical reactivity of the host strata. The deposits may be nonpersistent and highly variable in size, shape, grade, and texture of the ore. Typical ore is more coarse grained than syngenetic ore and may include local replacement of original framework grains. The associated metals are the same as found in syngenetic deposits. Favorable host rocks include nonmarine arkosic sandstones. The "red-bed" copper deposits near Cuba, N. Mex., are an example of

this kind of deposit. The other types of deposits have probably been modified to some degree by the processes that form this type of deposit.

THE KUPFERSCHIEFER

The Permian Kupferschiefer of Mansfeld, in the German Democratic Republic, is a thin-bedded bituminous marine marl or shale about 50 centimeters thick containing copper, lead, zinc, and silver in economic concentrations and many other metals in unusually large amounts (Richter-Bernburg, 1941; Wedepohl, 1964, 1971). The equivalent Zechstein shales of Lower Silesia in Poland contain economic concentrations of copper in some areas and of lead in other areas (Haranczyk, 1970). Many Europeans regard the mineralization of the Kupferschiefer and equivalent units as syngenetic because the mineralization is confined to a thin stratigraphic unit and its distribution is related to lithologic facies (Richter-Bernburg, 1951). Most advocates of the syngenetic hypothesis also suggest some extraordinary way of concentrating greater than usual amounts of copper from sea water. Submarine mineral springs and erosion of older copper ores are among the sources suggested for the copper. More recently, Brongersma-Sanders (1968) suggested that heavy metals could be concentrated in black shales and phosphorites by upwelling nutrient-rich waters. Davidson (1964) rejected all the arguments for unusual environments of deposition in favor of an epigenetic episode of mineralization by saline ground waters derived from the Zechstein salt deposits overlying the Kupferschiefer, citing as an example the metal-rich thermal ground waters encountered in wells near the Salton Sea as potential mineralizing solutions. Wedepohl (1971, p. 272) suggested a correlation between the local metal content of the Kupferschiefer and the soluble trace metals in the Lower Permian red sandstones over which the Kupferschiefer sea transgressed.

Much of the Kupferschiefer in West Germany is probably of submarginal grade, but there may be as much as several tens of millions of tons of rock with an average grade of 0.3 percent copper.

THE AFRICAN COPPER BELT

Nearly one-fifth of the world's production of copper comes from deposits in upper Precambrian sedimentary rocks that extend for about 300 miles across parts of Zambia and the Katanga province of Zaïre in Africa (U.S. Bur. Mines, 1970). These deposits are usually called "strata-bound copper deposits" because of some uncertainty as to their origin. Individual beds containing disseminated cop-

per sulfide minerals are persistent for many miles. The ore occurs in dark shales of the lower Roan Formation of the Katanga Supergroup. Most of the lower Roan Formation is made up of arkosic conglomerates and eolian sandstones overlying a granitic basement, but the dark shales of the main ore horizon also contain algal reefs. Thus, the ore horizon was probably deposited in a shallow restricted basin surrounded by algal reefs. An important feature suggestive of a syngenetic origin is the zonal arrangement of ore minerals within the ore formation parallel to the shoreline as it migrated back and forth with transgressions and regressions of the sea (Garlick, 1961, p. 155-160). The sequence proceeds from barren sediment along the shoreline through disseminated chalcocite, bornite, and chalcopyrite to pyrite in the same direction as the general flow of currents, as determined from crossbedding. However, the mineralized zone cuts stratigraphically upward through the succession from conglomerate to sandstone to argillite to impure dolomite.

The later discovery by Malan (1964) that chalcopyrite occurs in narrow argillaceous interstices between stromatolite columns in algal reefs is also cited as evidence that copper was deposited at the time of sediment deposition, perhaps by sulfate-reducing bacteria, or as a colloidal sulfate gel. Metal deposition in the copper, iron, and cobalt-bearing Chibuluma West ore body, Zambia, was syngenetic with sands and muds, according to Whyte and Green (1971) and could have been by (1) biogenic deposition of sulfides, (2) precipitation especially of iron and cobalt hydroxides by changes of pH where river waters entered the lake or branch of the sea, (3) detrital deposition of metalliferous minerals, or (4) by a combination of these methods. Many of the deposits in Zaïre have been altered by supergene enrichment, and they produce secondary copper minerals and a variety of other elements from dolomitic beds.

The very large resource of copper in the African copper belt may result from a combination of several important processes. If we assume that the deposits are syngenetic, they may have formed during the interval of Precambrian time when oxygen first became available in sufficient quantity to mobilize copper. Moreover, these deposits are at or near the base of a marine transgression which could mobilize copper in the underlying crystalline and nonmarine rocks in a manner suggested by Wedepohl (1971, p. 272) for the Kupferschiefer. Common features of the African copper belt and the Kupferschiefer were discussed by Putzer (1972).

The current rate of copper production from the African copper belt would suggest a known reserve of well over a billion tons of ore. Speculative resources may be several times as great.

PRECAMBRIAN SEDIMENTARY COPPER DEPOSITS IN THE UNITED STATES

At the White Pine district in Ontonogan County, Mich., copper occurs in the Nonesuch Shale of Precambrian Y age, especially in the basal parting shale bed (Ensign and others, 1968). About 5 percent of the U.S. copper production comes from this single bed which is 1-8 meters thick and contains anomalous copper over a distance of at least 250 kilometers (160 miles) in the Upper Peninsula of Michigan and northern Wisconsin. Chalcocite is the principal sulfide mineral in the mineralized zone whereas pyrite is the characteristic sulfide in the upper part of the Nonesuch Shale. The richest copper occurs in the darkest carbonaceous siltstones, shales, and sandstones, where chalcocite grains ranging in size from 2 to 20 microns are disseminated interstitial to the framework grains. Although the mineralization is persistent in certain beds throughout the district, a crosscutting relation between ore and bedding on a regional scale and the absence of a characteristic copper-bearing lithologic facies distinguish this deposit from the African copper belt. Consequently, an epigenetic, but not magmatic, source for the copper seems most likely. However, Robert Ehrlich and T. A. Vogel stated (oral presentation at Institute on Lake Superior Geology, Houghton, Mich., May 4, 1972) that according to their interpretation, the copper in the Nonesuch Shale is syngenetic, with some diagenetic modification of the distribution.

Strata-bound sedimentary copper deposits occur in Precambrian Y rocks of the Belt Supergroup in western Montana and adjacent parts of Idaho. Copper sulfides occur as disseminations, blebs, and veinlets in persistent beds of white fine-grained quartzite and siltite of the Revett Formation along a belt parallel to a broad domal feature formed before the close of Belt time. Anomalous amounts of copper are more widely known in green strata throughout the Belt basin. Most of the Belt rocks are feldspathic and contain variable amounts of mica (illite and sericite) plus chlorite in amounts inversely related to grain size. Anomalous amounts of silver and mercury are known to be associated with copper locally, but lead is known only in separate beds. The apparent relation between mineralization and a late Precambrian broad domal structure is suggestive of a diagenetic or early epigenetic

origin for the copper (Harrison, 1972). Later recrystallization and metamorphism to the biotite zone reduced rock permeability and precludes the possibility of late epigenetic mineralization.

Rocks of the Belt Supergroup, including the Revett Formation, may contain as much as 1 billion tons of ore with an average grade of 0.5–1.0 percent copper. Submarginal grade rocks with anomalous copper may be 10–100 times more abundant. Silver will be a valuable byproduct of the Belt ores when commercial exploitation begins.

PRECAMBRIAN SEDIMENTARY COPPER DEPOSITS IN THE U.S.S.R.

With an estimated reserve of at least 10 million tons of metallic copper (Sutulov, 1972, p. 10), the undeveloped sedimentary ores of Udokan in eastern Siberia (near 57° N., 117° E.) form one of the world's largest sedimentary copper deposits; Shank (1972, p. 88) reported that it may be the largest. The ores are mainly sedimentary and related to lithology, and the greatest ore concentrations occur in the 1,800–2,500-meter-thick Sakukan Formation within the 10,000–12,000 m Udokan Group of fine sandstone and siltstone and some thin lenses of carbonate (Bakun and others, 1958). The main layer within which ore masses occur is about 300 m thick, its top 250 m below the top of the Sakukan Formation. The top of a lesser ore-bearing layer is 1,250–1,950 m stratigraphically beneath the main one. Despite the great thickness of the Udokan Series, deposition was in shallow water, or in a littoral and deltaic environment.

The Udokan Series is considered by most geologists to be of Precambrian X age, but some geologists think it as young as Precambrian Z. Copper minerals were deposited with the sedimentary material or were diagenetic (Bogdanov, 1968). The copper is believed to have come from a Precambrian W shield area located to the northeast, containing anomalous quantities of copper. Part is believed to have been transported as fine clastic particles.

The Udokan Series is folded into a big synclinal basin, irregularly oval in plan; the southwest limb is overturned, but folding is not complex. The entire series is slightly metamorphosed, and some of the rocks are now quartzites and phyllites.

The main ores are disseminated chalcocite and bornite; there are also some chalcopyrite ores that are not associated with the preceding. Malachite, brochantite, and other secondary minerals are present in a restricted oxidized zone. Sphalerite, native silver, and molybdenite are rare in the deposit. Ore has been traced to depths of more than 1,500 m.

Lyubimov (1966, p. 231) described the remote location and problems that face development in the Udokan Series. The region has severe climate, permafrost, and seismicity and an elevation of more than 1,000 m above sea level, but the high metal content of the ore and the possibility of open-pit mining probably will compensate for the high capital expenditures needed.

RED-BED COPPER DEPOSITS

Numerous deposits of copper are associated with red sandstone strata of late Paleozoic or early Mesozoic age, as described by Lindgren (1911). In the Southwestern United States, only a few of the deposits are large enough to be mined economically. Host rock strata, interbedded with red strata, are generally light gray to dark gray and may weather yellowish gray. Nonmarine feldspathic or arkosic sandstones containing carbonaceous matter are the most common host rocks. Such copper deposits are similar in size and shape to some of the well-known deposits of uranium and vanadium in sandstone, and transitional deposits containing all three metals exist. The copper deposits generally contain some silver, and at Silver Reef, Utah, silver values exceed those for copper.

Near Cuba, N. Mex., the Nacimiento copper deposit, which occurs in the Agua Zarca Sandstone Member of the Chinle Formation, is a typical large red-bed copper deposit (Kaufman and others, 1972). Rich ore occurs as chalcocite, bornite, covellite, and secondary copper minerals, replacing fossil wood and forming halos around the wood. The deposit is clearly epigenetic, but appears to predate the major structural features of the area. In past years selective mining of rich chalcocite replacements of logs left much disseminated ore in the ground. The remaining disseminated and rich copper ore may be as much as several million tons in this area alone. The total speculative resources of copper in red-bed copper deposits of the United States may be several hundreds of millions of tons of mineralized material.

A second type of copper deposit generally included with red-bed copper deposits is represented by disseminated copper sulfide minerals currently mined from thin beds of gray shale in the Flowerpot Shale of Permian age in southern Kansas and western Oklahoma (Ham and Johnson, 1963; Johnson and Ham, 1972). The copper-bearing shales occur in a sequence that includes red sandstone, dolomite, gypsum, and shale and are facies equivalents of evaporite deposits including salt in western Kansas and in the Midland and Delaware basins of

Texas and New Mexico. Like the Kupferschiefer of Europe, these deposits have lateral persistence within thin strata, which is commonly cited as evidence for a syngenetic origin.

Strata-bound copper-bearing sandstones that occur in Carboniferous rocks of Dzhezkazgan in central Kazakhstan, U.S.S.R., are included here even though descriptions of these deposits indicate significant differences from our red-bed copper deposits. Copper sulfides with some lead occur in gray sandstone horizons interbedded with barren brown sandstones of shallow marine lagoonal and deltaic facies. As many as 27 zones of mineralized sandstone occur in 700 m of enclosing rock. The gray host rock is more calcareous and dolomitic than the associated barren rock. The primary ore consists of sandstone impregnated with bornite, chalcocite, chalcopyrite, and galena concordant with the lamination of the sandstone. The first stage of mineralization is probably syngenetic or diagenetic with the enclosing strata; it is cut by a second, epigenetic stage of copper-lead-zinc mineralization that forms intersecting veins, stockwork zones, and hydrothermal wallrock alteration. The ore-forming solutions may have been derived from basaltic magma at depth below the sea floor according to Shcherba (1971). These deposits were estimated to have a reserve of more than 3.5 million tons copper metal in ore containing 1.6 percent copper (Sutulov, 1967, p. 113).

MODERN SEA FLOOR NODULES

Extensive deposits of ferromanganese nodules now forming on the deep-sea floor in the major oceanic basins are examples of syngenetic copper-bearing deposits that may become economic in the near future as technology progresses. Although the origin and mode of concentration of metals in these nodules are the subject of intense debate, there can be no question that the nodules represent syngenetic deposition. The copper content of the nodules varies regionally in the ocean basins; the higher concentrations are normally on the west sides of the basins and between lats 10° N. and 10° S., 2,500–6,000 m below the surface. The nodules in the richer areas contain more than 25 percent manganese, as much as about 1.5 percent copper and nickel, and 0.25 percent cobalt, as well as many other elements in smaller quantity. The copper ions are thought to be adsorbed on the manganese oxides; copper minerals have not been reported. These nodules are discussed in more detail in "Manganese"; Mero (1965) reported information as to distribution and quantity.

METAL-RICH SEDIMENTS OF THE RED SEA

In March 1964, during an oceanographic survey of the Red Sea, scientists aboard the RRS *Discovery* found hot brine in the Discovery Deep. Subsequent exploration has shown that underlying the hot brine are iron-rich, brine-saturated sediments at least 10 m thick containing anomalous amounts of copper and zinc. The geology, chemistry, and physical properties of these sediments are discussed in detail in a collection of reports edited by Degens and Ross (1969). Aside from the economic potential of these deposits (Bischoff and Manheim (1969) estimated 50 million tons of brine-free sediment averaging 1.3 percent copper), the greatest interest lies in the theoretical implications of these deposits with respect to the origin of sedimentary deposits elsewhere. James (1969) suggested that deep burial of deposits of this type and continued circulation of hot brine through them would result in re-solution of the base metals and lateral migration. He speculated that such migration would continue until the solutions reached the H₂S-bearing reef limestones known to be laterally continuous with the metal-rich sediments, there to precipitate as sulfide minerals. Such a model provides a basis for new appraisals of the theories of origin of many types of sedimentary base-metal deposits. Furthermore, since the hot brines which have introduced the metals to the sea floor appear to be related to continental rifting, a tectonic control for sedimentary deposits is suggested. If stratigraphic and paleotectonic studies of a region suggest rifting and deep basin filling, favorable host rocks along the flanks of the rift may contain deposits of base-metal sulfides.

HORIZONTAL TRANSPORT OF SECONDARY COPPER DEPOSITS

A few localities are known where copper dissolved from sulfide-bearing rock as a part of the conventional surficial leaching has then traveled laterally and been deposited in a secondary zone that is not located above the primary sulfide. Perhaps this phenomenon is more common than is suspected, and consideration might be given to searching for such ores in the vicinity of subeconomic sulfide mineralization.

Lateral transport and secondary deposition of copper has been described at the Emerald Isle mine, Wallapai mining district, Arizona (Dings, 1951, p. 149–153), Jerome, Ariz. (Lindgren, 1926, p. 90–91), Chuquicamata, Chile (Newberg, 1967), Globe-Miami district, Arizona (Throop and Buseck, 1971), and Ray, Ariz. (Phillips and others, 1971). The transport distance at Chuquicamata was about 4

kilometers. Most of the deposits are in Quaternary gravels, but some are in older rocks.

MOBILIZED DEPOSITS OF PROBABLY SEDIMENTARY ORIGIN

Two major deposits, Aitik in northern Sweden and Mount Isa in Queensland, Australia, contain copper ores that were probably sedimentary in origin but have been modified by metamorphism. The ores of the old White Pine mine, Michigan, were deposited by hydrothermal solutions, but the source of the metal was probably the nearby strata-bound copper present in the Nonesuch Shale, the ore of the present mine. Many other deposits may also fall into this general category, and classification is harder as modification has increased.

The large low-grade copper deposit of Aitik, near Gällivare, Sweden, is in Precambrian gneisses, schists, and quartzites (Zweifel, 1972). The copper is in chalcopyrite and occurs with pyrite, magnetite, and pyrrhotite, mainly as disseminations and stringers. There was some hydrothermal alteration along with the mobilization of the sulfides.

At Mount Isa, lead-zinc-silver ores form masses that are conformable with the bedding of the Precambrian Mount Isa Shale, but the generally associated copper ores are in shears, particularly where silica and dolomite have been introduced. Thus, the lead-zinc-silver ores are reasonably interpreted as syngenetic, but the origin of the copper sulfide ores is less certain.

ENVIRONMENTS OF DEPOSITION

An understanding of the origin and environment of deposition of sedimentary copper deposits has practical application in the search for new or hidden ore deposits. A search conducted in depositional environments comparable to those of known copper deposits should have a far greater chance for success than a random search. Environments worth consideration include depositional environments where syngenetic concentrations are optimum, as well as postdepositional environments where diagenetic or epigenetic concentration seem most likely, as follows:

1. Deep marine basin sediments associated with midocean ridges or continental plate boundaries, such as the recent sediments along the crest of the East Pacific rise and sediments in association with the hot brine pools of the Red Sea.
2. Continental-margin marine black shales associated with phosphorites, such as the Meade Peak Phosphatic Shale Member of the Phosphoria Formation of Permian age in western Wyoming.
3. Shallow marine or estuarine shale associated with evaporites or nonmarine deposits, such as the Flowerpot Shale of Permian age in Oklahoma and the Kupferschiefer of Permian age in Germany and the equivalent Zechstein beds in Poland.
4. Sandstone, shale, and carbonate rocks exposed to migrating saline ground waters, such as alkaline carbonated waters or chloride brines derived from subsurface solution of evaporites. Examples might be the "Speckled Sandstone" of probable Early Permian age in the Salt Range of Pakistan, the Flowerpot Shale of Permian age in Oklahoma, the Santa Rosa Sandstone of Triassic age in New Mexico, and the San Andres Limestone of Permian age in New Mexico.
5. First-cycle sedimentary rocks in a restricted or closed basin, such as the Agua Zarca Sandstone Member of the Chinle Formation of Triassic age along the west flank of the Nacimiento Range, N. Mex.
6. Pyritic shale associated with submarine volcanism, as perhaps the Nonesuch Shale of Precambrian Y age in Michigan and Wisconsin.
7. Pyroclastic or porous carbonate-bearing rocks exposed to ground water from weathering copper-sulfide deposits, like the Apache Leap Tuff of Miocene age at the Inspiration mine, Arizona (Throop and Busek, 1971).
8. Stratified rocks of mostly Precambrian Y age (800–1,600 million years), especially if they are first-cycle sediments deposited in shallow-water or restricted basin environments. Examples are the Revett Formation of the Belt Supergroup in northwestern Montana, and the Roan Formation of the Katanga Supergroup in the African copper belt.
9. Basal transgressive marine rocks directly overlying deeply weathered crystalline rocks or first-cycle arkosic clastic strata, such as the Kupferschiefer of Permian age in central Europe and the Roan Formation of Precambrian age in the African copper belt.
10. Porous strata updip and marginal to deeply buried volcanic rocks of basaltic to intermediate composition, as the amygdaloidal flow tops and conglomerate beds of the Portage Lake Volcanics and perhaps also the Nonesuch Shale, both of Precambrian Y age in the Upper Peninsula of Michigan.

11. Porous or chemically reactive strata, including lava flows and pyroclastic rocks, within or peripheral to rocks of the zeolite facies of low-grade metamorphism. Examples besides the Nonesuch Shale are the Sangre de Cristo Formation of Pennsylvanian and Permian age in southern Colorado and the Apache Leap Tuff at the Inspiration mine, Arizona.

The environments described above are not all mutually exclusive; therefore, a coincidence of several may greatly increase the probability of finding significant copper mineralization.

SIZE AND GRADE CHARACTERISTICS

Sedimentary deposits are difficult to characterize as to size. On the one hand, many separate mines are not in separate ore deposits but in parts of a continuously mineralized stratum, and on the other hand, very large tonnage figures have been mentioned for beds continuously mineralized over wide areas even though only a small part may be mined economically. In figure 22, a field of seven deposits in Zaïre and Zambia has an average copper content of 4 million tons each in ore averaging 3.5 percent copper. Totals of production and reserves for both the White Pine and Keweenaw Point districts in Michigan amount to 7 or 8 million tons of metal for each deposit but in lower grade ore. Estimates for the Udokan deposit in the U.S.S.R. exceed 10 million tons contained copper.

MASSIVE SULFIDE DEPOSITS IN VOLCANIC ROCKS

Copper is more abundant in basalt and andesite than in other igneous rocks. Under various conditions, this copper content may be concentrated to form an ore deposit of distinctly different form from those associated with felsic intrusions.

A small but locally important part of the world's copper resources is found in lavas and pyroclastic rocks laid down in the marine environment. These deposits are generally (1) stratiform or lenticular, and concordant with the bedding of the surrounding rocks, (2) of small lateral extent as compared to thickness, and (3) composed largely of sulfide minerals with small proportions of silicate gangue. Mineralogically the deposits consist mainly of pyrite and (or) pyrrhotite and varying amounts of chalcopyrite, sphalerite, and galena. The terms "massive sulfide," "cupriferous pyrite," or "massive pyrite" deposits have been used to identify this class.

The last decade has been extremely productive in theories of origin of massive sulfide deposits. Recog-

nition of metamorphic features in the North American deposits by Kinkel (1962) and Kalliokoski (1965) permitted the interpretation that their emplacement was earlier than the events that deformed them. Regional geologic studies such as those of Goodwin (1965) in the Noranda area, Quebec, made possible the views that the deposits were a part of the volcanic rock complexes that enclosed them and not a later superimposed feature. Detailed studies of Kuroko-type ores in Miocene tuffs of Japan (for example, Horikoshi, 1969) showed that these ores were formed on the sea floor during the last stages of an eruptive cycle. The equivalence of these deposits to the metamorphosed massive sulfides of older terranes had by this time become clear to many workers.

Anderson (1969) presented a thorough review of the genesis of these deposits, and in 1970 the meeting of the International Mineralogical Association-International Association on the Genesis of Ore Deposits in Japan brought together an outstanding group of papers on the subject. The description which follows is largely taken from papers by Tsumi and Watanabe (1971), Hutchinson and Searle (1971), and L. A. Clark (1971) in the proceedings of that meeting.

Massive sulfide deposits that are undeformed by later orogenic activity exhibit a wide variety of textures and forms. Ore may be found in disseminations and stockworks or swarms of veins in highly propylitized lava. Above these may be found sulfide ore in cavities between pillows as at Cyprus. Massive sulfide bodies above the stockworks and vein swarms may show banding and colloform texture indicating deposition in open cavities or on the open sea floor. The ore may be intensely brecciated with interfragmental spaces filled with fine-grained sandy pyrite, and evidence exists for gravitational transport of blocks of ore, lava, and soft bottom sediment downslope on the sea floor from the source of the deposit. Overlying the massive ore are bedded deposits rich in iron oxides, silica, and disseminated pyrite. These bedded deposits are thickest over the massive ore, and they thin abruptly laterally.

The preceding features have given rise to the volcanic exhalative theory of origin. Metal-bearing solutions or vapors are thought to ascend rapidly through fractures in a submarine volcanic pile and to precipitate, partly in colloidal form, in near-surface cavities and on the sea floor. Deposition of sulfides may temporarily seal the vent area, and the resulting increase of vapor pressure may cause the brecciation of previously deposited ore. Mixing of iron-rich solutions with oxygen-rich circulating

sea water results in deposition of the overlying bedded siliceous ironstones.

The source of metals for these deposits is the same magma that gives rise to the surrounding volcanic rocks. Different lava types give rise to deposits of different mineralogy. For example, pyrite-pyrrhotite-chalcopyrite deposits are commonly associated with mafic volcanics, and sphalerite- and galena-bearing deposits tend to be found with silicic volcanic rocks (Anderson, 1969).

Copper-bearing massive sulfide deposits contain as much as 10 percent copper, with a mean in commercial ores of about 2.5 percent, and they range in size from a few hundred thousand tons to more than 50 million tons. The wide range in production grade of copper is due in part to the variable content of zinc, lead, silver, and gold in these ores as well as the economics of recovery of sulfur and iron from them.

COPPER IN NICKEL ORES

Many of the commercial nickel deposits in mafic rocks throughout the world have copper associated with them, and this "byproduct copper" is an important contribution to the supply, especially in Canada and the U.S.S.R. These deposits are generally interpreted to have been formed by a magmatic-segregation process as a part of the intrusion of the bodies of mafic rock.

The chalcopyrite-pentlandite ores related to mafic intrusives range in composition from mostly copper sulfides to mostly nickel sulfides, depending on the composition of the primary magma; the proportion of nickel increases roughly with maficity (T. P. Thayer, oral commun., 1972). In certain cumulate mafic bodies this may result in high-nickel sulfide at the base and more cupreous sulfide in higher strata (Thayer, 1971, p. 242). In these ores the sulfide has generally separated as an immiscible liquid, and the richest concentrations in these sulfur-rich magmas formed when the sulfide separated before crystallization had gone very far. Many papers in Wilson (1969) deal with the origin of ores of this type.

The most significant producer of copper from nickel-bearing mafic rocks is the Sudbury district of Ontario. In this area of complex geology, the ores, some stratiform and some crosscutting, are polygenic in relation to their enclosing rocks but were probably all derived initially as products of crystallization and the separation of a sulfide-rich dense liquid. Variation in the Ni:Cu ratio in different ore bodies makes it possible for a mining company to adapt the ratio in the ore produced to

metal-market conditions.

Besides the Sudbury district, nickel sulfide deposits containing appreciable quantities of copper are found as segregations in mafic intrusive bodies in Maine and New Brunswick, in Quebec, Manitoba, Saskatchewan, and particularly in the coastal crystalline belt of British Columbia and southeast Alaska. The copper-nickel ores of Norway, Sweden, Finland, and the Kola Peninsula, U.S.S.R., are mostly polygenic and complex. The nickel-copper-cobalt ores of Noril'sk in north-central Siberia are also an important source of copper for the U.S.S.R. Nickel ores in Precambrian ultramafic and mafic rocks in the newly discovered Kambalda district in Western Australia contain 0.25–0.57 percent copper.

The deposits in Minnesota are in the Precambrian Duluth Gabbro Complex in the northern part of the State. Disseminated copper and nickel sulfides form discontinuous lenses and layers in the basal part of the gabbro complex. Recent studies show that although these deposits are not minable under present conditions, they represent a very important future copper resource (P. K. Sims, oral commun., 1972).

NATIVE COPPER DEPOSITS OF THE KEWEENAW TYPE

On the Keweenaw Peninsula, Mich., deposits of native copper occur in amygdaloidal flow tops and conglomerate beds in the Portage Lake Volcanics. These strata-bound and in part conglomerate-bed-controlled ores are mostly interpreted to be epigenetic in origin and may have affinities to some strata-bound ores in sedimentary rocks. (Bateman (1950, p. 498) likened them to the sandstone ores of Corocoro, Bolivia.) The proximity in both space and time of ores in the Portage Lake rocks to the mineralized Nonesuch Shale, 350–6,000 feet (110–1,850 m) above, also suggests this affinity.

Opposing this reasoning are the widespread minor copper occurrences in basaltic rocks all over the world, suggesting that the copper was originally syngenetic but has been redistributed and concentrated to minable grades by later processes (Cornwall, 1956, p. 620).

More than 5 million tons of copper metal has been produced on the Keweenaw Peninsula since mining began in 1845, and although there is little activity in the district now, large ore reserves remain.

Most of the copper is the native metal and was introduced into vesicles, fractures, and interstitial pore spaces after deformation of the rocks. Moreover, the copper deposits fit into a pattern of low-grade regional metamorphic zoning of silicate min-

erals similar to the zeolite facies described by Coombs, Ellis, Fyfe, and Taylor (1959). The source of mineralizing solutions for the copper sulfide deposits in the Nonesuch Shale and the native copper deposits in the Portage Lake Volcanics may be essentially the same—namely, the 15,000–30,000 feet of flood basalts at depth in the Lake Superior Basin (Ensign and others, 1968, p. 486). The copper-bearing solutions, according to White (1968, p. 322), may represent water driven from the voids and interstices of flow tops as the lava flows were crushed and metamorphosed in the axial region of the Lake Superior syncline, downdip from the present deposits.

Other copper occurrences that resemble the Keweenaw type are the chalcocite-bearing Coppermine deposits in gently dipping basalt flows of Precambrian Y age in the Northwest Territories, Canada, and the native copper-cuprite-azurite-malachite ores in chloritized and epidotized mafic flows in the Catoclin Formation (Precambrian Z) of Maryland and Virginia. Chilean copper deposits in terrestrial volcanic rocks such as Buena Esperanza (Ruiz and others, 1971) may be similar in origin to the Keweenaw type, although they lack native copper. Bornite, chalcocite, and chalcopyrite are found in amygdaloidal upper parts of andesite flows in these deposits.

MISCELLANEOUS TYPES

Some copper deposits of local and (or) historic importance differ significantly in form and rock association from those of the five major groups just described. Among these are small and very high grade chalcocite deposits such as Kennecott, Alaska, and Mountain City, Nev. (See fig. 22 for comparison with other types.) Nearby volcanic rocks have been called upon as a source of copper in these deposits (Coats and Stephens, 1968; E. M. MacKevett, written commun., 1972). The massive chalcopyrite lenses and veins of Matahambre, Cuba, appear to be unrelated to either volcanic or intrusive rocks (R. A. Bradley, oral commun., 1972). The sulfide lenses in shears crosscutting anorthosite bodies at Chibougamau, Quebec, do not lend themselves to inclusion with any of the previously described types.

Replacement ore bodies in carbonate rocks far from an igneous source, such as at Bornite-Ruby Creek, Alaska (Runnels, 1969), may be an important resource of copper. Numerous lodes of this type in a 450-mile belt along the south slope of the Brooks Range, Alaska, are believed to contain more than 10 million tons of copper metal. Small copper production (12,000 tons in 1970) from Missouri is

a byproduct of the lead-zinc replacement bodies in carbonate rocks without igneous affiliation (Allen Heyl, oral commun., 1972).

RESOURCES

IDENTIFIED AND HYPOTHETICAL RESOURCES

About 70 percent of the world's identified copper resources fall into four distinct geologic-geographic groups. These are, in decreasing importance, porphyry copper deposits of Chile and Peru, porphyry copper deposits of the Southwestern United States, sedimentary copper deposits of Zaire and Zambia, and porphyry copper and sedimentary copper deposits of the U.S.S.R. Other important groups are the porphyry copper deposits in Oceania, in Mexico, and in western Canada and the porphyry copper and sedimentary copper deposits in southeastern and central Europe, respectively. Volcanogenic deposits and nickel-copper deposits in mafic intrusions, important as they may be to local economies, do not contribute significantly to the worldwide picture.

Identified copper resources, given in table 39, are

TABLE 39.—*Identified and hypothetical copper resources, in millions of short tons*

Area	Identified ¹	Hypothetical ²
United States:		
Eastern -----	10	5
Western, except Alaska ---	64	75
Alaska -----	2	20
Canada -----	19	50
Mexico -----	18	20
Central America -----	1	6
Antilles -----	2	1
South America -----	80	50
Europe, excluding U.S.S.R. ----	25	20
Africa -----	53	50
U.S.S.R. -----	39	50
Middle East-south Asia -----	4	20
China -----	3	?
Oceania, including Japan -----	21	30
Australia -----	3	3
Total -----	344	400

¹ Identified resources: Specific, identified mineral deposits that may or may not be evaluated as to extent and grade and whose contained minerals may or may not be profitably recoverable with existing technology and economic conditions. Based on all categories of reserve figures plus estimates where no figures are available. Amounts are tentative and accuracy will be refined in subsequent publications.

² Hypothetical resources: Undiscovered mineral deposits, whether of recoverable or subeconomic grade, that are geologically predictable as existing in known districts. Based generally on identified resource figures times a factor assigned according to geologic favorability of the region, extent of geologic mapping, and exploration.

those for which a size estimate is available. All categories of reserve figures as well as estimates for deposits where no reserve figures are published have been combined to make up this identified resource. These figures should be regarded as tentative, and considerable refinement of their accuracy should be expected in subsequent publications. They are derived from unpublished resource compilations made by A. R. Kinkel and others in the early 1960's, with additions and subtractions based on new dis-

coveries and estimated mine production, respectively. Low-grade deposits are not included here but are discussed under "Subeconomic Resources" of this chapter.

Hypothetical resources given in table 39 are those that have not been found but whose existence is suspected, as in concealed parts of known mineral districts, in projections of incompletely explored mineral belts, and in areas near important isolated new discoveries that suggest the possibility of a future mining district. The largest hypothetical reserves are in the Western United States, South America, Africa, and the U.S.S.R., all areas of favorable geologic setting and much recent successful exploration.

Hypothetical resources in the Western United States comprise the concealed deposits under basin fill in the major porphyry copper regions, a possible new porphyry copper province in the Absaroka Range of Wyoming, and the great resources of sedimentary copper in the Belt Supergroup of Idaho and Montana. Alaskan hypothetical resources are based on favorable evaluation of porphyry deposits in eastern Alaska and of replacement deposits in carbonate rocks on the south slope of the Brooks Range. Canadian hypothetical resources are in British Columbia and Yukon Territory. South American resources are in the Andes of Chile, Peru, Ecuador, and Argentina, and the estimate is probably conservative. African hypothetical resources are in the copper belt region and in the Precambrian rocks of southwest Africa. U.S.S.R. hypothetical resources are large, and the estimate of 50 million tons may be conservative; the resources comprise undiscovered porphyry deposits in the Kazakhstan and Uzbekistan regions and copper-bearing sedimentary strata in several areas. Another area where future discovery of important copper porphyry deposits can be reasonably expected is in the central Iranian volcanic belt northwest from the known porphyry occurrences near Kerman through the area near Hamadan and Tabriz and then westward across northern Turkey. In this area some types of copper mineralization related to intrusions have been reported by Superceanu. (1971, p. 395). The areas of copper prospects in northeastern and east-central Iran that may have potential for large copper deposits are discussed under "Speculative Resources."

The identified resources in the United States, 76 million tons of copper metal, are enough to last the country 45 years at the constant-use rate of 1.7 million tons per year; the world identified resource of 344 million tons will last 50 years at the rate of 6.9 million tons per year. At the same use rates,

hypothetical resources add 60 years supply to the United States and 58 years supply to the world.

SPECULATIVE RESOURCES

During the last decade many new mineral provinces and districts were discovered and developed. Important examples of these are the porphyry copper deposits of the Canadian cordillera, the Southwest Pacific, and Iran. These areas had previously received little attention, and deposits of the porphyry type were thought to be limited mostly to Southwestern North America and the Andes. Similarly, sedimentary copper deposits, previously considered as rather unique occurrences, have been discovered in the Western United States and in eastern U.S.S.R. The development of these new provinces has drastically increased the resources.

There is considerable likelihood of finding even more new copper provinces given the new knowledge of copper metallogeny and perhaps aided by the same combination of economic boom and high metal prices that existed in the last decade. Future discoveries of porphyry copper and sedimentary copper deposits are more likely to have a major impact on the world's copper resources, and these types will receive the greatest attention. Unfortunately, not enough is known of the stratigraphic and paleotectonic controls of sedimentary copper deposits to be able to point to many areas with potential resources of this type.

In the United States, the greatest likelihood of new discoveries lies in the Southwest. In Arizona, the search for new porphyry copper deposits has been concentrated in the southeastern part of the State because of the greater number of known deposits and because more is known about the geology, geophysics, and geochemistry. Most of the southwestern, western, and northwestern parts of the Basin and Range province has not been evaluated adequately in reconnaissance, much less in detail. Adequate geologic maps exist for only a few widely separated areas. The reasons for this are many and include inaccessibility, restricted access to military bombing ranges and Indian reservations, and inhospitable climate. Nevertheless, from what is known, these are pristine exploration areas, and must be considered to contain many copper deposits. On the basis of a simple comparison with the concentration of copper deposits in southeastern Arizona, and generally favorable regional geology, an order of magnitude of 10^8 tons is estimated for these areas.

Elsewhere in the Western United States, the Mountain City copper deposit in Elko County, Nev.,

which yielded over 100,000 tons of copper in ore averaging nearly 10 percent, is believed to have been derived from the nearby metavolcanic rocks of the Nelson Formation of Late Mississippian age (Coats and Stephens 1968). Volcanic rocks of Mississippian to Permian age are widespread in northwestern Nevada, and these and nearby sedimentary rocks offer numerous prospecting targets for high-grade copper deposits (Hewitt, 1968).

Metal-rich brines in the Salton Sea area of southern California and the Boleo copper deposits in Pliocene rocks in Baja California, Mexico, suggest a prospecting area. Upper Tertiary to Holocene rocks and sediments in and along both sides of the Gulf of California rift zone and its northern projection in the Salton trough may have been or may be in the process of being mineralized by metal-rich solutions rising through this rift zone.

In the Appalachian Mountains, the possibility of finding large porphyry copper deposits of Paleozoic age is intriguing. Ore-bearing porphyries are believed to be emplaced only at relatively high levels in the crust, during the final or postorogenic stages of a tectonic cycle. Thus pre-Mesozoic fold belts contain few of these deposits, because erosion in these areas has progressed to the point where most of the upper part of the crust has been removed. Special conditions, however, such as the deep burial and preservation of rocks of an early tectonic cycle by geosynclinal deposits of a later cycle, may result in the preservation of some older porphyry-type deposits such as at Catheart Mountain in Maine and those of the U.S.S.R. in central Asia.

Guild (1971) and Sillitoe (1972) discussed the relationship between porphyry copper deposits and the boundaries of major crustal plates and concluded that many deposits are related to the process of magma generation along the subduction zone where one crustal plate slides beneath another. Porphyry copper deposits tend to occur along the concave side of arc-trench systems resulting from the process of subduction. Recognition of ancient crustal-plate boundaries of this type would enable us to outline episubduction regions where porphyry copper deposits might occur. The Piedmont Province of the Appalachian Mountains may be such a region. H. L. James ("The Metallogenic Concept," Presidential address. Soc. Economic Geologists, Nov. 13, 1970, unpub. rept.) pointed out that this province has an arrangement of ore-bearing zones suggesting and eastward-dipping subduction zone. If porphyry copper deposits still are preserved they would lie in the Piedmont near the present edge of the Coastal Plain. Copper, lead, and zinc have been

noted in several old mines forming a trend extending 400 miles northeastward from central Georgia to the North Carolina-Virginia boundary. Within this trend, several low-grade disseminated copper-molybdenum occurrences have been explored by drilling programs in recent years. Porphyritic intrusive rock has been reported from only the Magruder mine area in Lincoln County, Ga., but geologic reports on the old gold mines and prospects are mostly sketchy and limited to descriptions of the veins and the closely associated rocks. Some of the mines and prospects are reported to have areas of abundant disseminated pyrite.

In southwestern Alaska and the Aleutian Islands, geologic conditions similar to those of the Southwest Pacific permit the possibility of discovery of porphyry copper deposits. Quartz diorite intrusions and wide areas of disseminated pyrite described on Amchitka Island (Powers and others, 1960, p. 539, 541) suggest that in at least one locality a favorable environment exists for porphyry copper prospecting. Projection of the porphyry copper provinces of British Columbia and Yukon Territory into central Alaska suggests that large resources might be found there.

In the Antilles, an extensive copper province is suggested by the occurrence of porphyry copper deposits in Puerto Rico. The Puerto Rican deposits were formed during a period of vulcanism and intrusion in the early Tertiary (Barabas, 1971) and are overlapped by middle and upper Tertiary coastal plain carbonate rocks. A small molybdenum prospect in an Eocene pluton in the Virgin Islands (Ratte, 1971) suggests an eastward extension of this mineral province. In the limestone Caribbees, lower Tertiary plutonic rocks are exposed on St. Martin and Antigua. Hydrothermally altered rocks in St. Martin, Guadeloupe, Martinique, and other islands suggest interesting drilling targets, some of which were being explored by private companies during 1971. It is possible that another million tons of copper might be discovered in these islands, but land-use conflicts between mining and tourism may make their recovery in the near future doubtful.

West of Puerto Rico the lower Tertiary mineral province is interrupted. Volcanic rocks and potentially productive intrusive bodies are absent in the lower Tertiary sequence of Santo Domingo. However in the Montagnes Noires of Haiti, the Blue Mountains of Jamaica, and the southern Oriente Province of Cuba, lower Tertiary volcanic rocks are present. Quartz-bearing intrusive rocks and alteration zones that may be present in these areas represent high-priority targets for porphyry copper

exploration.

In western Colombia, quartz diorite intrusions of probable middle Tertiary age in the Sierra Occidental (Irving, 1971) suggest a large region where porphyry deposits might be found. Recent geochemical investigations by U.S. Geological Survey and Colombian government geologists have revealed strong copper anomalies in one of these intrusions in the north end of the Sierra (E. M. Irving, oral commun., 1972).

In Panama, recently discovered porphyry copper deposits related to upper Tertiary volcanic rocks (Ferencic, 1970; Dengo and others, 1969) suggest a wide province with potential mineralization.

Economically important copper deposits are unknown in the Precambrian shields of South America, but the region is vast, nearly unpopulated, and difficult of access. In Brazil, the recent highway construction and westward pioneering movement and accompanying geologic and geochemical exploration may yield copper resources of great size. Not enough is known of the geology of this region to do more than speculate on the possibility of major sedimentary copper deposits of the Zaire-Zambia type or smaller deposits of the Tsumeb type.

In western Europe and northwest Africa, porphyry copper deposits of late Paleozoic age may be preserved. These could be products of a Paleozoic subduction zone resulting from the collision of the European-African and the American crustal plates. Goossens (1971) suggested the possibility of porphyry copper occurrences in western Africa.

The Alborz Mountains of northern Iran, with numerous small copper occurrences, may be a long extension of the copper-molybdenum porphyry province in Soviet Armenia. The areas of volcano-sedimentary complex northeast of the Dasht-e-Lut in eastern Iran and in the northern part of the Chagai District in Pakistan, both areas of some late volcanism, are favorable areas for the occurrence of very young (Pliocene-Pleistocene) porphyries. One untested porphyry at Saindak in the Chagai District was described (Schmidt, 1968), and one prospect in east-central Iran, that at Gazu, was described in a way suggesting a porphyrylike deposit (Bazin and Hübner, 1969, p. 107-108).

Mesozoic fold belts of the Soviet Far East probably contain large porphyry copper deposits. The Koryak Peninsula would be the logical beginning for the search for these because of the apparent continuity of cordilleran-type structures into this area. Younger deposits may occur along the Kurile-Kamchatka-Aleutian arc-trench system.

Recent discovery of porphyry copper deposits in

the Philippines, Solomon Islands, and Borneo suggests that all the island arc areas in the Western Pacific, especially the Indonesian islands, might be potential prospecting ground.

A region of abundant felsic volcanic rocks in northeastern China and eastern Mongolia would seem to have potential for significant copper-porphyry and massive sulfide deposits. Other ranges besides the Ch'i-lien-shan, especially in northern Tsinghai Province, might contain significant massive sulfide copper deposits. There is also considerable potential for sedimentary copper deposits in many parts of this large country. We do not know enough about recent Chinese geologic literature to speculate on extensions of the environment of the copper porphyry of Chung-t'iao-shan in Shansi Province.

The Tasman geosynclinal belt in eastern Australia contains many copper occurrences and major deposits of the massive sulfide type. Late Paleozoic felsic intrusions in this belt may contain large resources of the porphyry copper type.

Table 40 summarizes our estimate of the amount

TABLE 40.—*Summary of speculative resources, in millions of tons, of copper¹*

Basin and Range porphyry copper	100
Alaska porphyry copper	10
Other Western United States possibilities, Appalachian Mountains porphyry copper, and Antilles porphyry copper	10
Central America—Colombia porphyry copper---	10
Brazilian shield sedimentary copper	50
Western Europe-northwest Africa porphyry copper	10
Middle East porphyry copper	10
U.S.S.R. Far East and Kurile-Kamchatka arc--	100
China	(?)
Australia	10
Indonesia	10
Total	320

¹ Speculative resources: Undiscovered mineral deposits, whether of recoverable or subeconomic grade, that may exist in unknown districts or in unrecognized or unconventional form. Based mostly on estimates of geologic favorability and extent of geologic mapping.

of copper that might be contained in these untapped districts and provinces. The figures are at best a guess and may be in error by as much as 100 percent.

SUBECONOMIC RESOURCES

Lowell (1970) estimated that in the upper mile of the earth's crust there are about 3×10^{15} tons of copper. The recoverable identified, hypothetical, and speculative ore deposits discussed in the previous section make up a small fraction of this amount. Another fraction, subeconomic resources, comprise bodies of mineralized rock containing copper in amounts considerably higher than crustal abun-

dance but lower than could be mined under present economic and technological conditions. Subeconomic resources may be either identified or hypothetical, but inasmuch as data on grade and tonnage of subeconomic deposits are rarely available, no distinction is made between these categories. Subeconomic speculative resources represent such a high degree of uncertainty that they cannot be given numerical values. The lower limit of copper content of deposits in the subeconomic category depends on one's personal expectations of revolutionary advances in mineral recovery and mining methods. Copper will ultimately be mined at lower grades than most other metals because recovery is facilitated by its tendency to occur as minute sulfide grains rather than in silicates. However, adverse effects on the environment and the magnitude of the labor and power requirements mitigate against the creation of huge open-pit mines in rocks containing only a few hundred parts per million of copper.

Table 41 gives some of the identified and hypo-

TABLE 41.—*Partial list of subeconomic (conditional) resources, in millions of tons, of copper*¹

Locality	Type of deposit	Copper resources
Western United States -----	Porphyry copper -----	20
Do -----	Sedimentary -----	20?
Eastern United States -----	Sedimentary -----	11
Do -----	Copper-nickel -----	60
Sea floor -----	Nodules -----	50 (40-60)
Western Canada -----	Porphyry copper -----	20
South America, Andes -----	do -----	100
Africa -----	Sedimentary -----	50
Europe -----	do -----	50
Total -----	-----	381

¹ Conditional resources: Specific, identified mineral deposits whose contained minerals are not profitably recoverable with existing technology and economic conditions.

thetical subeconomic resources that may become economically important in the future.

In the United States, conditional resources in porphyry copper deposits are probably very large. Lowell (1970, p. 69) predicted that average grades in open-pit copper mines will decline to 0.25 percent by the year 2000. In-place leaching of deposits of even lower grade may soon become economical. In addition, some large deposits such as at Safford, Ariz., are too deep or too irregularly distributed to permit profitable mining at the present time but may become minable in the near future. An estimate of the hypothetical subeconomic resources in porphyry copper deposits averaging 0.25 percent copper requires some knowledge of the frequency of such deposits compared with deposits of minable grade. Moreover, to predict the increase in reserves of known deposits in response to a drop in average ore grades requires some knowledge of the rate at which copper content falls off with distance outward

from boundaries of porphyry copper ore bodies. If the slope of the curve of copper plotted against distance outward is steep for most deposits, decreases in average grade will have little effect on reserves. Inasmuch as these data are not available, we are reduced to making the simple assumption that there are at least as many deposits averaging 0.1-0.4 percent as there are deposits of currently minable grade. The discovery and development of this additional 7-8 billion tons of material in the western porphyry copper provinces with average grades of about 0.25 percent would yield about 20 million tons of additional copper.

Large conditional resources in sedimentary copper deposits in the United States are difficult to evaluate. Geologically, sedimentary deposits pass from economic ores to subeconomic resources when the beds in which the copper minerals occur become too thin or become separated by thick beds of barren rock. Also, some beds may have engineering characteristics that make mining dangerous or expensive, or a steeply dipping copper-bearing bed may reach depths that are beyond the reach of present-day underground mining practice. Nevertheless, the copper in certain sedimentary deposits is known to be continuous at fairly constant grade for great distances.

The Nonesuch Shale at White Pine, Mich., is such a deposit. W. S. White (oral commun., 1972) estimated that 11 million tons of copper metal exists in currently subeconomic mineralized beds. The sedimentary copper deposits recently being explored in the Belt Supergroup of Montana and Idaho probably contain large conditional resources, but no information is yet available on their minable reserves.

The copper-nickel deposits in the Duluth Gabbro Complex in Minnesota represent a large conditional resource. P. K. Sims (oral commun., 1972) estimated that 14 billion tons of mineralized rock exists along the lower contact of the gabbro body. This material has a lower cutoff of 0.25 percent and an average grade of 0.58 percent combined copper nickel. Assuming a 3 to 1 copper-nickel ratio, these deposits contain slightly more than 60 million tons of copper metal. Some of this copper is disseminated in bodies as large as 100 million tons and may soon become amenable to mass-mining methods, according to Sims.

Summing up these large conditional resources (table 41) gives a total of 111 million tons of copper which may at some time become available in the United States.

Among the world's most important conditional resources are those of the sea floor. These comprise

copper-bearing manganese nodules of the ocean basins and brine-associated metal-rich bottom sediments of the Red Sea. On the basis of Mero's (1965) estimates of copper content and abundance of ocean floor manganese nodules (see also "Manganese"), we estimate that 40–60 million tons of copper may be available from this source. Development of these resources depends on the accurate determination of their size, research and development on economical mining and metallurgical methods, and settlement of the international-legal problems of ownership.

Elsewhere in the world, reasonable assumptions on which to base estimates of presumed resources become increasingly difficult to make. In the Andes, where porphyry copper deposits are large and are mined at much higher grade than those in the United States and Canada, there must exist very large subeconomic resources. The sedimentary deposits of Zaïre and Zambia probably also contain very large amounts of copper in thin or deeply buried beds. A doubling of the copper reserves of these areas is not difficult to imagine when modern low-cost mining methods are established and (or) world copper prices rise.

Wedepohl (1971) gave 30–100 million tons as the probable copper content of the Kupferschiefer in Germany. Economic deposits are largely depleted, and this resource lies in thin or deeply buried beds.

PROSPECTING TECHNIQUES

We must assume that most easy-to-find copper deposits in the populated regions of the world have now been discovered. For all except some remote areas, new discoveries are more difficult to make and indirect methods of prospecting are being used increasingly to find copper and other metals. From now on, our success in finding new and especially hidden deposits will greatly depend on our ability to develop new methods of detecting concealed ores.

Some prospecting techniques can be applied to a wide variety of ores, but others are only for specific targets. Geochemical reconnaissance is often the first step in prospecting and usually seeks several metals that may be dispersed from any kind of ore deposit. Stream sediment sampling, described by Hawkes and Webb (1962), is the most effective reconnaissance method in many terranes. Refinements of this method, such as improved statistical analysis of data and sampling of certain mineral species (for example, magnetite), have had some success. Detailed study of anomalous areas and outlining of drilling targets are usually accomplished by soil sampling and analysis. In tropical areas copper can be mostly leached from soils overlying the deposits,

but Learned and Boissen (1972) showed that over porphyry copper deposits, traces of gold associated with chalcopyrite remain in the soil and that the deposit may be outlined by the gold anomaly over it. In glaciated regions, ore fragments in boulder trains have been studied in hope of tracing them to their source; sulfide fragments were followed for 50 km in the discovery of the Outokumpu ores of eastern Finland.

Rocks containing widely disseminated pyrite associated with copper porphyry deposits, in combination with other effects of hydrothermal alteration, weather in the desert environment to produce a large color anomaly, "distinguished by brownish to greenish coloration, bleaching and variegated iron staining" (Jerome, 1966, p. 82–83). In some localities large red-brown areas ("red thumbs") have developed near the ore bodies as at Bisbee, San Manuel, and Banner-Pima-Mission, all in Arizona. Recognition of the "red thumb" area from an aircraft played an important part in the exploration of the La Caridad deposit, Mexico. The areas of color anomaly may be hundreds of meters to several kilometers in size and might be recognizable on colored aerial photography (Schmidt, 1968, p. 60). They also form target areas large enough to be sought from instrumented satellites; tests of the detection of a porphyry ore body in a desert area are planned for the Earth Resources Technology Satellite A.

Preliminary experiments have been made to compare the spectral reflectance from trees in soil over a sulfide mineral deposit with reflectance from trees of the same species on nearby unmineralized soil (Canney, 1970). Early results of tests made on a disseminated copper-molybdenum deposit in Maine have been encouraging.

Geophysical methods of many types are suitable for copper prospecting, but they are much more specific for certain types of ore. Aeromagnetic surveys are used to outline bodies of mafic rocks which may contain massive sulfide or magmatic-segregation deposits or to indicate areas of weakly magnetic hydrothermal alteration or highly magnetic skarn mineralization associated with porphyry deposits. The use of geophysical methods in the search for copper porphyry deposits was discussed by Brant (1966) and Sumner (1970). The potassic alteration that accompanies porphyry copper deposits can, theoretically, be detectable by modern airborne gamma-radiation surveys (Moxham and others, 1965). Electromagnetic surveys by ground or air are used in outlining massive sulfide deposits; induced-potential surveys are effective in locating

porphyries and other types of rocks that contain disseminated sulfides. The use of geophysical methods in the search for massive sulfide ores is discussed by several authors in Morley (1970).

Recent developments in geochemistry and geophysics have made it possible to recover more information from diamond drilling than simply the geology and copper assays of the core. Cheaper and rapid analytical methods, principally in spectroscopy, have made it more feasible to analyze large numbers of samples for many elements. Slight variation in trace-element composition may be caused by metals dispersed outward from an ore body not intersected by the hole. Similarly, down-hole geophysical instruments of many types have been perfected, enabling the geologist to extend his knowledge many feet beyond the walls of the drill hole.

PROBLEMS FOR RESEARCH

Figure 22 shows the relationship between the size and grade of ore deposits and copper production. Annual U.S. production is about equivalent to a deposit the size of Sierrita in Arizona, and annual world production is about equal to a La Caridad deposit or an average Zaïre-Zambian sedimentary deposit. It is important to know if we are mining our copper deposits faster than we are finding them. During the last 10 years, U.S. mine production was about 13 million tons of copper (table 36). New porphyry copper deposits discovered and (or) brought into production during this 10-year period, listed by Peters (1970), will yield slightly more than 14 million tons of copper. Assuming that several million tons of copper is known in other, less important types of deposits and in some large but unannounced discoveries, it appears that, in the United States, exploration and development are ahead of production by a small margin. Worldwide exploration and development lead mine production by at least 28 percent, again using Peters' list of discoveries.

The last decade was notable for intense exploration activity in response to favorable political conditions and high copper prices. Moreover, economic geologists experienced a renaissance in geologic theory during this period, in which new approaches to problems of geochemistry, igneous and sedimentary petrology, and tectonics were found and applied to ore deposits. However exploration geology is very sensitive to economic cycles, and there is reason to believe that this period of activity is over, at least temporarily (Colen, 1972). This stop-and-start exploration activity is extremely wasteful of human

energy, and if we are to satisfy the long-term demand for metal in a future world in which ore deposits become increasingly more difficult to find, new ways of organizing exploration activities must be sought, especially long-range planning.

Copper is a commodity for which a permanent worldwide shortage is not easily foreseeable, but even though the hypothetical and speculative resources may be very large, an enormous amount of geologic field work, analysis, and even intuition will be required to transform these resources into proven reserves. At the same time we must continue to develop our scientific knowledge in order to replace outdated exploration ideas with new ones.

On the geological side of resource study, the need for geologic and geophysical maps as a base for mineral exploration cannot be overemphasized. More specifically, additional efforts must be applied to mapping and other geologic studies of those parts of the United States that have the highest mineral potential. Southeastern Arizona is probably the richest copper province in the world, yet geologic mapping at a scale of 1:62,500 is complete for only 27 percent of the area, and nearly half the province is unmapped.

Large-scale metallogenic maps might be an effective exploration tool in southeastern Arizona to detect the presence of deposits concealed beneath younger volcanic and sedimentary rocks.

Lowell and Guilbert (1970) and others pointed out that almost all porphyry copper deposits have a peripheral zone of small veins of base and precious metals, manganese carbonates, and barite that may extend outward for several miles from the deposit. Careful mapping of these veins, including even the smallest mineral occurrences, and careful classification of them according to various mineralogic, structural, textural, and genetic schemes may direct the exploration geologist to blind porphyry ore bodies or to those covered by later deposits. This mapping should be done at scales of 1:125,000 or greater depending on the complexity of the geology and spacing of the data points. Trace-element analyses of the veins and of mineral separates from them such as biotite or chlorite may aid in the interpretation. Studies of salinity and temperature of formation of fluid inclusions in the vein materials may show the geologist the direction toward the source of the mineralizing solutions.

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